

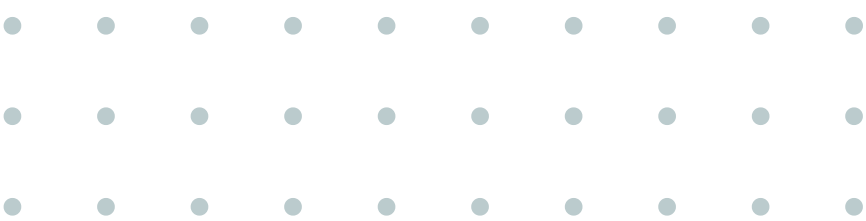
Clustering

Discovering structure in unlabeled data
By Muhammad Nanda S. / Data Scientist



Session Objectives

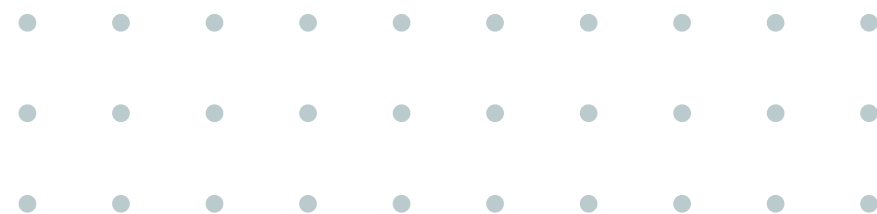
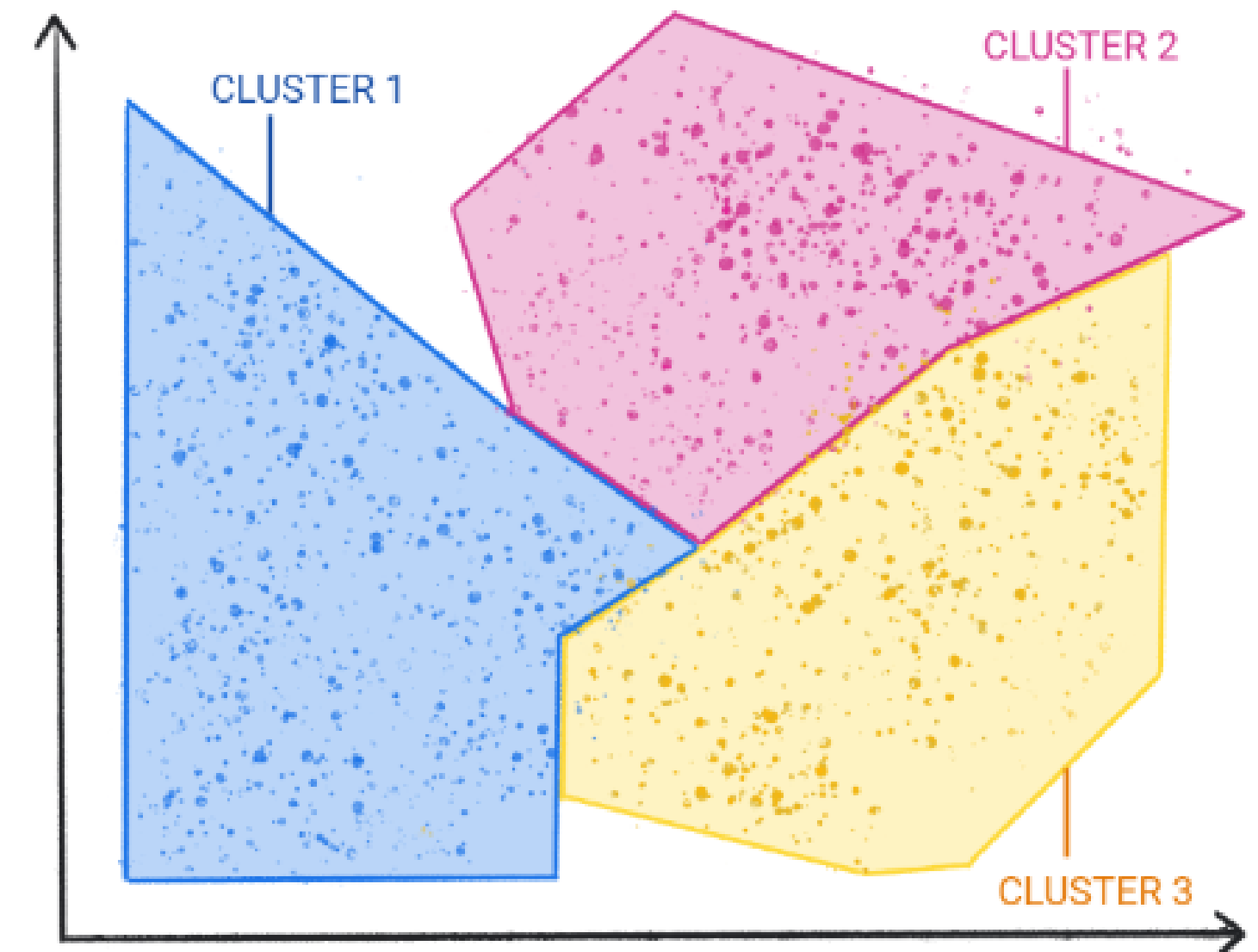
- 01. Grasp **core clustering concepts** and common algorithms.
- 02. Learn practical ways **to select and evaluate** cluster models.
- 03. See a real-world case and understand business actions from clusters.



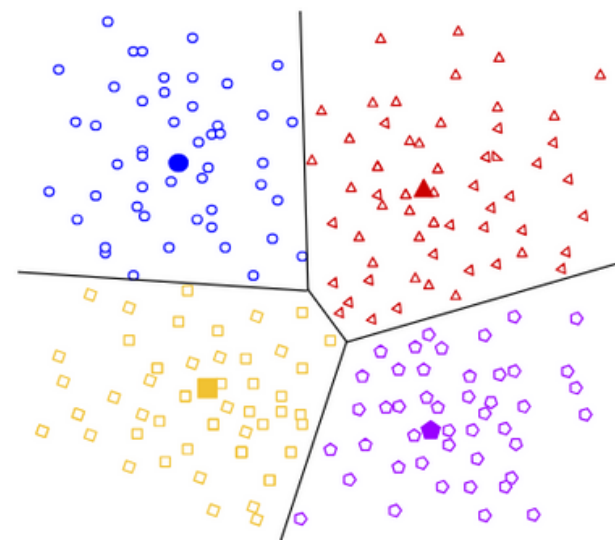
Clustering Algorithm

Grouping unlabeled examples into clusters based on meaningful patterns is called clustering.

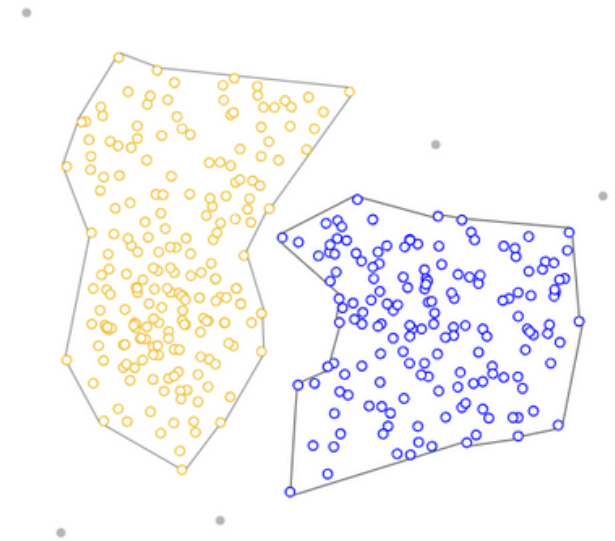
- Market segmentation
- Social network analysis
- Grouping similar news



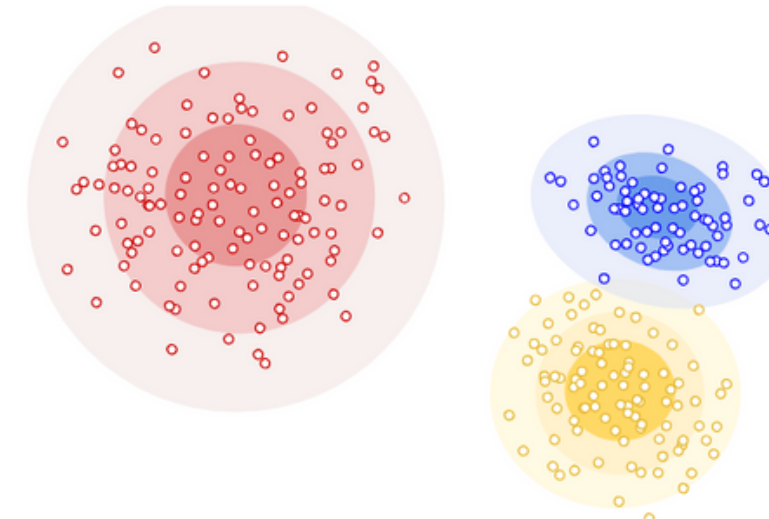
Type of Clustering Algorithm



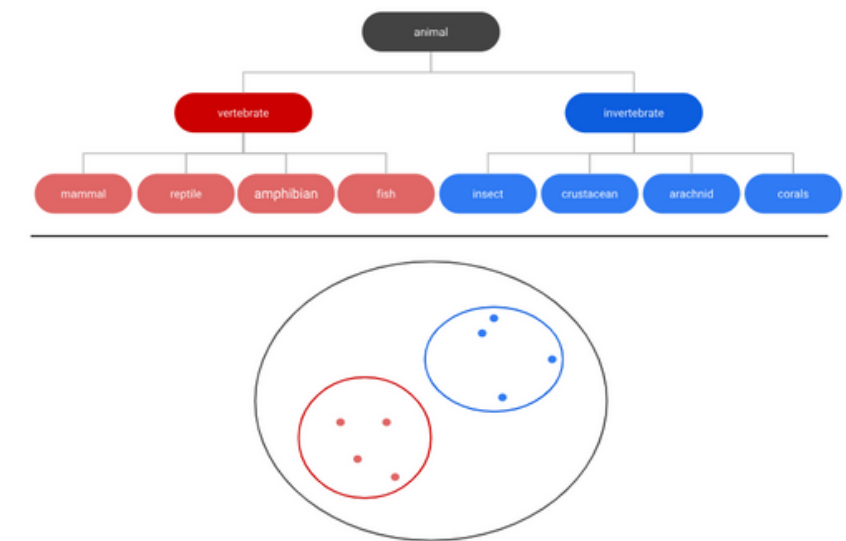
Centroid-based
Clustering



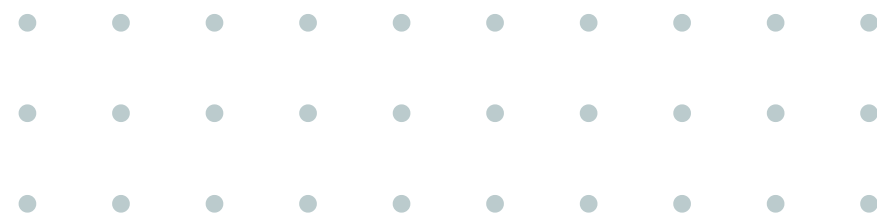
Density-based
Clustering



Distribution-based
Clustering

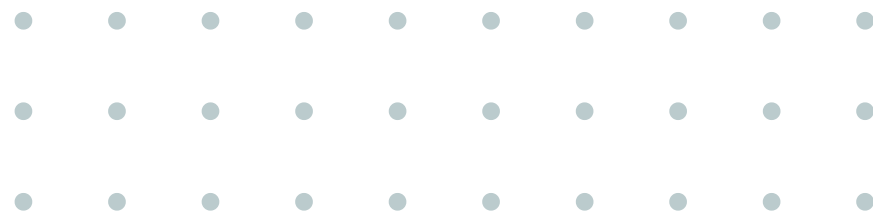
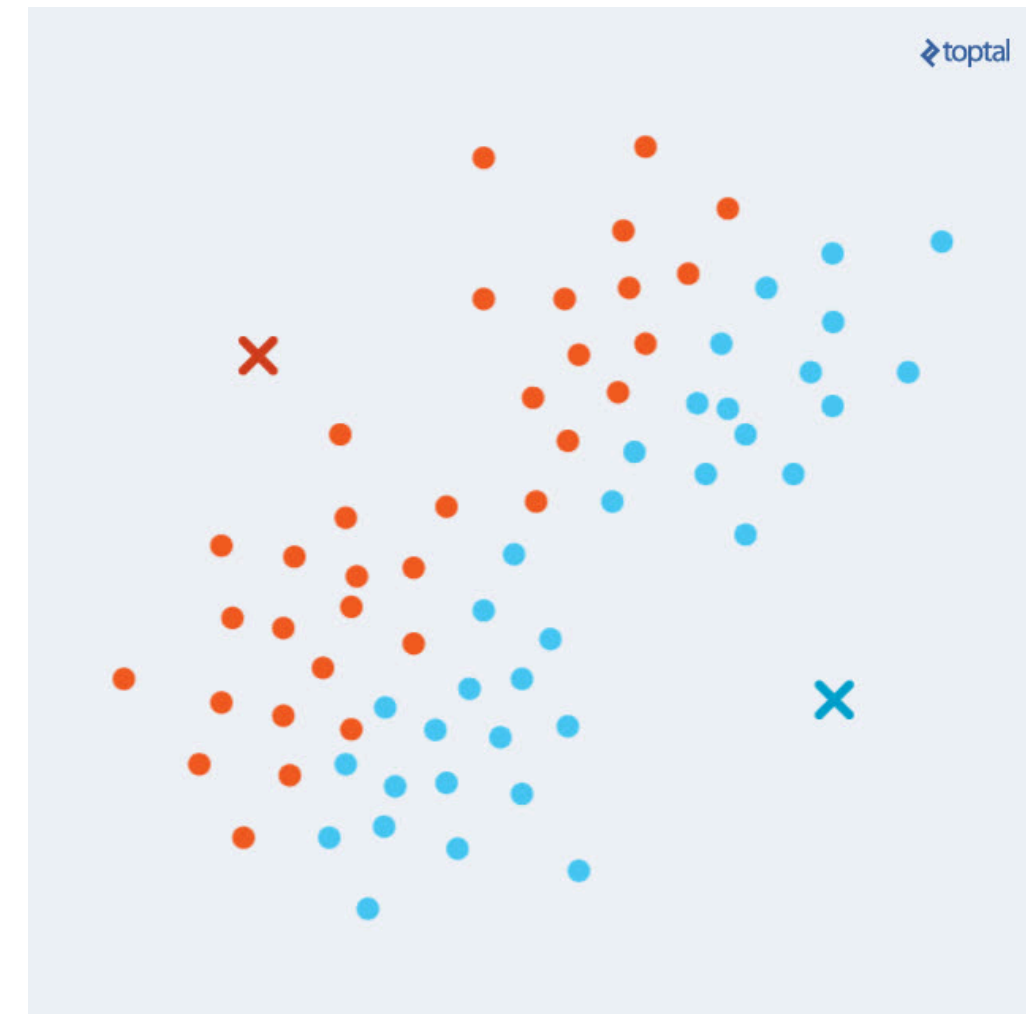


Hierarchical-based
Clustering



K-means Algorithm

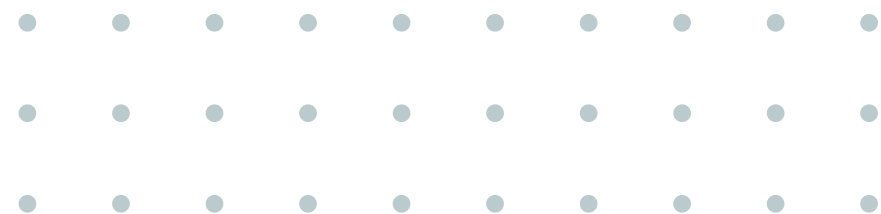
- Randomly **chooses K** cluster centroid.
- Assigns each point to the **closest centroid** to get K initial clusters.
- For every cluster, **recomputes the centroid** by taking the average (mean) of all points in the cluster.
- **Repeats** the calculation of centroids and assignment of points until the centroid points stop changing.



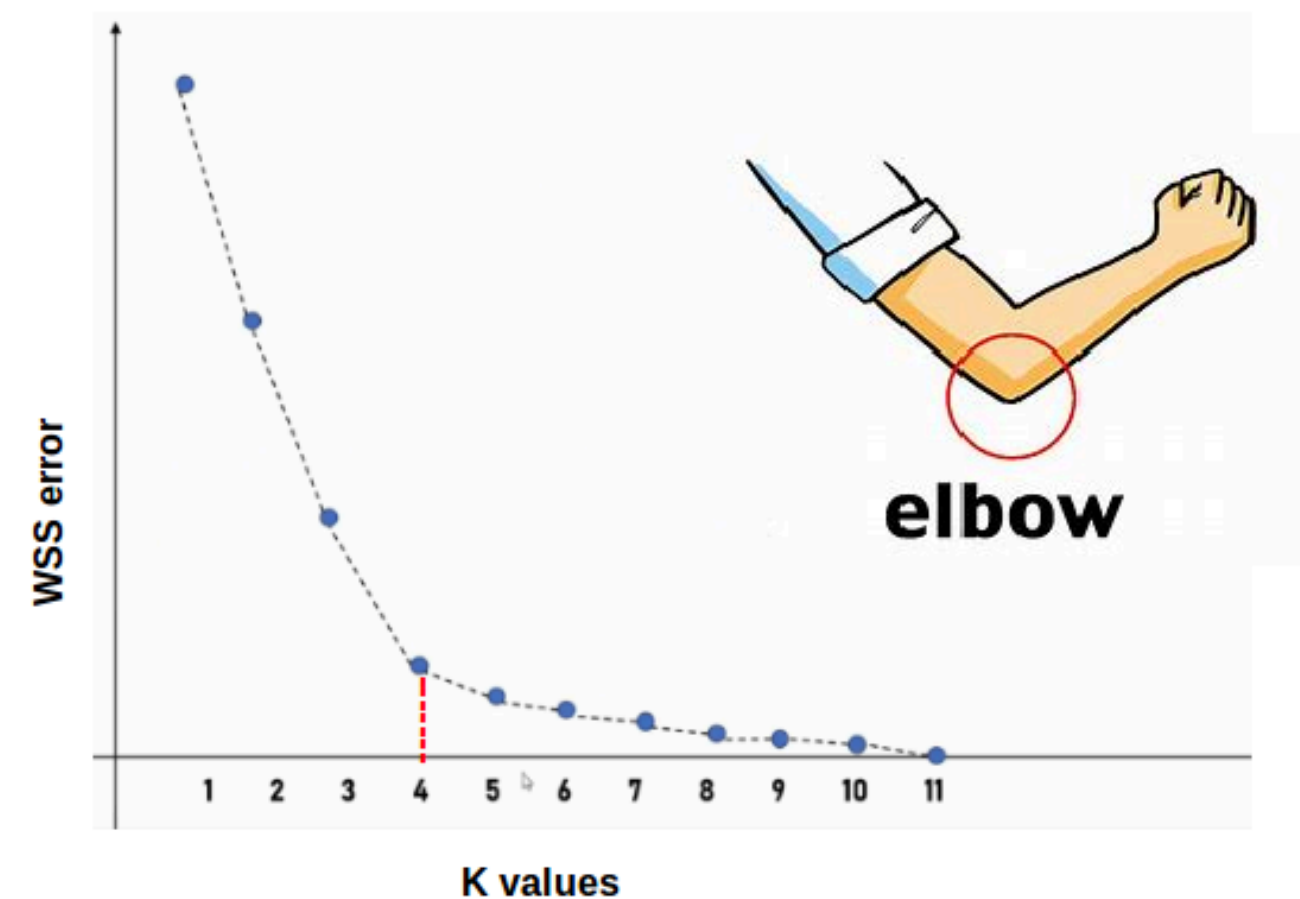
How to Choose the Number of Clusters

A. Elbow Method

- Inertia is calculated by summing the squared distances.
- The Elbow Method is one of the most popular methods to determine this optimal value of k .
- In this method, we chose the value of k for which there is a sharp and steep fall of the inertia.



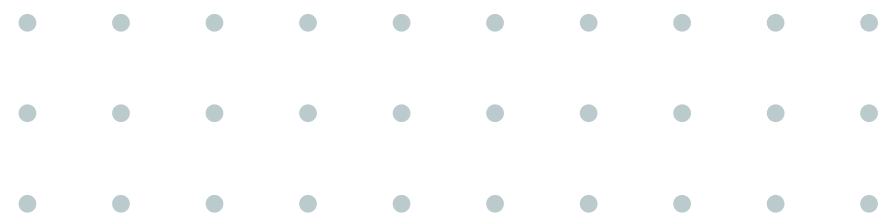
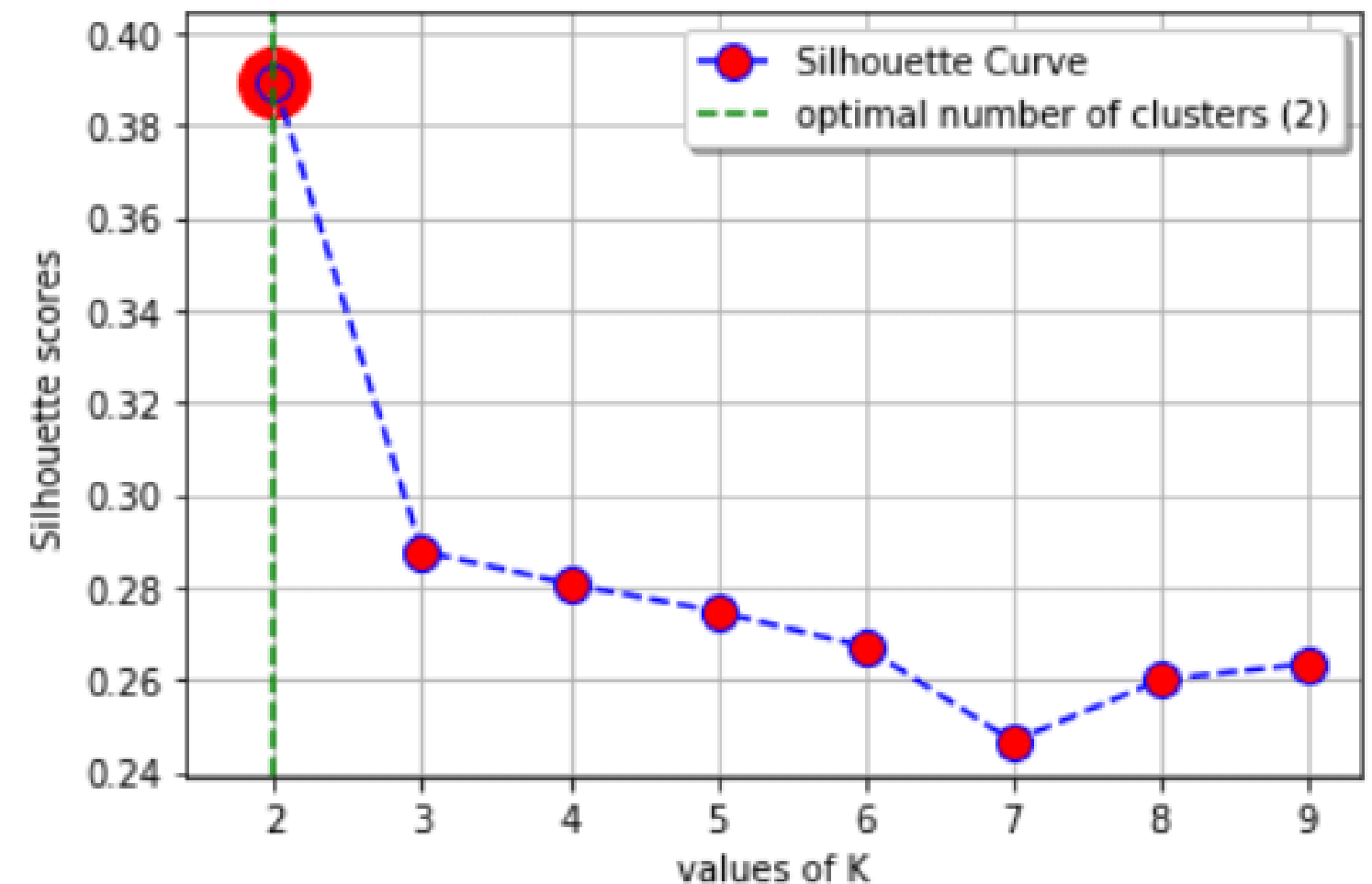
Elbow method



How to Choose the Number of Clusters

B. Silhouette Score

- Combines cohesion (how close points are to their own cluster) and separation (how far from other clusters)
- Score ranges from -1 to +1
 - +1 → well-clustered, far from others
 - 0 → overlapping or unclear clustering
 - -1 → likely misclassified
- The optimal k is the one that maximizes the average silhouette score



Advantages & Limitations

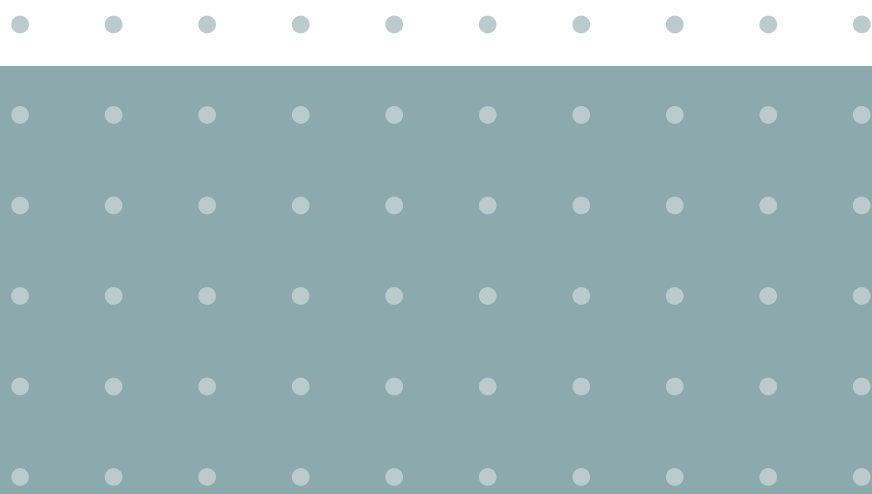
Advantages

- Reveals hidden structure in unlabeled data.
- Useful across many domains (marketing, social graphs, text).
- Fast and interpretable (for simple methods).

Limitations

- Requires choosing K (many algorithms sensitive to parameters).
- Some algorithms assume cluster shape.
- Sensitive to scaling and outliers (preprocessing is essential)





Let's Discuss!

Do you have any question?

